

A Perturbative Approach to the Ground State of Interacting Fermions in the Mean Field Limit

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Objective

Mathematically rigorous analysis of the ground state energy of interacting fermions in the limit $N \rightarrow \infty$.

Mean Field Limit

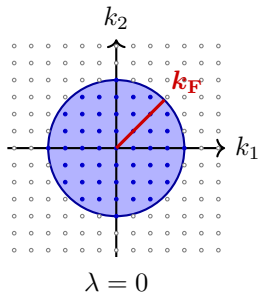
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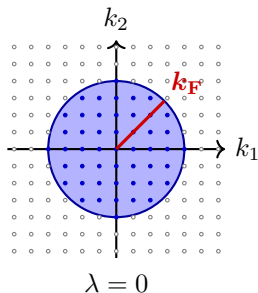


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 $\implies$ **kinetic energy** grows as $\mathcal{O}(N^{5/3})$;

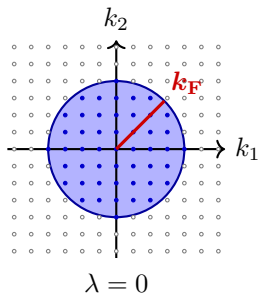


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 $\Rightarrow$ **kinetic energy** grows as $\mathcal{O}(N^{5/3})$;
- number of interactions is $N(N-1)/2$
 $\Rightarrow$ **interaction energy** grows as $\mathcal{O}(N^2)$ if V is of order $\mathcal{O}(1)$.

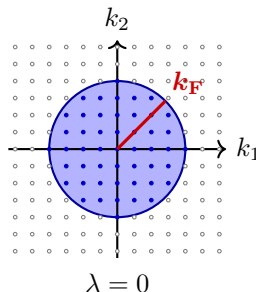


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Mean field scaling

To have an extensive H_N ($H_N \sim N$) in the limit $N \rightarrow \infty$, we define the **effective parameters**

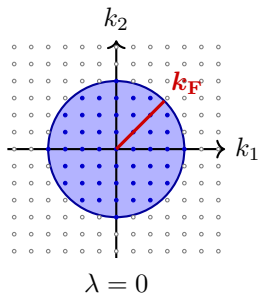
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Mean field scaling

To have a non-trivial limit $H_N / N \sim N$ in the limit $N \rightarrow \infty$, we need to scale the parameters as follows:

semiclassical parameter

mean field coupling constant

$$\hbar = N^{-1/3}$$

and

$$\lambda = N^{-1}$$

Main Result

With ordinary second order perturbation theory [Hainzl-Porta-Rexze, 2019] and then rigorous bosonization methods [Benedikter-et-al, 2021]:

$$E_N^{\text{gs}} = E_N^{\text{HF}} + E_N^{\text{RPA}} + o\left(N^{-1/3}\right) \quad \text{as } N \rightarrow \infty$$

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As $N \rightarrow \infty$ we have $E_N^{\text{gs}} = E_N^{\text{HF}} + o(1)$, and

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Long-term objective: To understand the connection between bosonization and QFT methods.

Sketch of the proof: definitions

Second quantized Hamiltonian:

$$H = \sum_{k \in \mathbb{Z}^3} (\hbar^2 |k|^2 - \mu) a_k^* a_k + \lambda \sum_{p, q, k \in \mathbb{Z}^3} \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p$$

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The **free energy** is $F = -\frac{\log(Z)}{\beta}$, and $E_N^{\text{gs}} = \lim_{\beta \rightarrow \infty} F$.

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Duhamel's formula at second order for $\log(Z)$:

$$\begin{aligned} \log \operatorname{Tr} e^{-\beta H} &= \log \operatorname{Tr} e^{-\beta H_0} - \lambda \beta \langle V \rangle_{\beta, H_0} \\ &\quad + \lambda^2 \int_0^\beta \int_0^\tau [\langle V(\tau') V(\tau) \rangle_{\beta, H_0} - \langle V(\tau') \rangle_{\beta, H_0} \langle V(\tau) \rangle_{\beta, H_0}] d\tau' d\tau \end{aligned}$$

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Wick's theorem: $H_0 = \sum_k \epsilon_k a_k^* a_k$ and $A_j = a_j^*, a_j$, then

$$\begin{aligned} \langle A_1 \cdots A_{2n} \rangle_{\beta, H_0} &= \sum_{\sigma \in P_{2n}} (-1)^\sigma \prod_{j=1}^{2n-1} \langle A_{\sigma(j)} A_{\sigma(j+1)} \rangle_{\beta, H_0} \\ \langle A_1 \cdots A_{2n-1} \rangle_{\beta, H_0} &= 0 \end{aligned}$$

where P_{2n} is the set of **pairings** of $2n$ elements.

Sketch of the proof: Feynman diagrams

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$$\sum_{p,q,k} \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p = \begin{array}{c} p-k \quad q+k \\ \diagdown \quad \diagup \\ \quad \bullet \\ \diagup \quad \diagdown \\ q \quad p \end{array}$$

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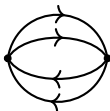
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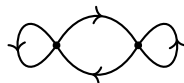
$$\begin{aligned} \lim_{\beta \rightarrow \infty} \lambda \langle V \rangle_{\beta, H_0} &= \lim_{\beta \rightarrow \infty} \lambda \left(\text{diagram 1} + \text{diagram 2} \right) \\ &= N^{-1} \sum_{\substack{k \in \mathbb{Z}^3, p \in B_F: \\ p-k \in B_F}} \left(\hat{V}(0) - \hat{V}(k) \right) \\ &= \mathcal{O}(N) + \mathcal{O}(1). \end{aligned}$$

Future perspective: a first look at second order

At second order, the **connected Feynman diagrams** are split into two possible topologies:



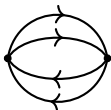
No tadpoles



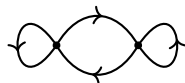
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However, they contain unphysical results, like the **tadpole divergence**:

$$\lim_{\beta \rightarrow \infty} \frac{\beta}{2} \left(\text{Diagram with two tadpoles and internal lines labeled } p-k, q+k, p', q', p, q, p'-k', q'+k' \right) = \lim_{\beta \rightarrow \infty} \left(-\frac{\beta}{2} \sum_{p,q,k} \hat{V}(0) n_q^2 n_p n_k \right) = -\infty$$

Conclusion

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Ultimate goal: to take the proof of the N -scaling of the ground state energy to all orders in perturbation theory.

Thank you!